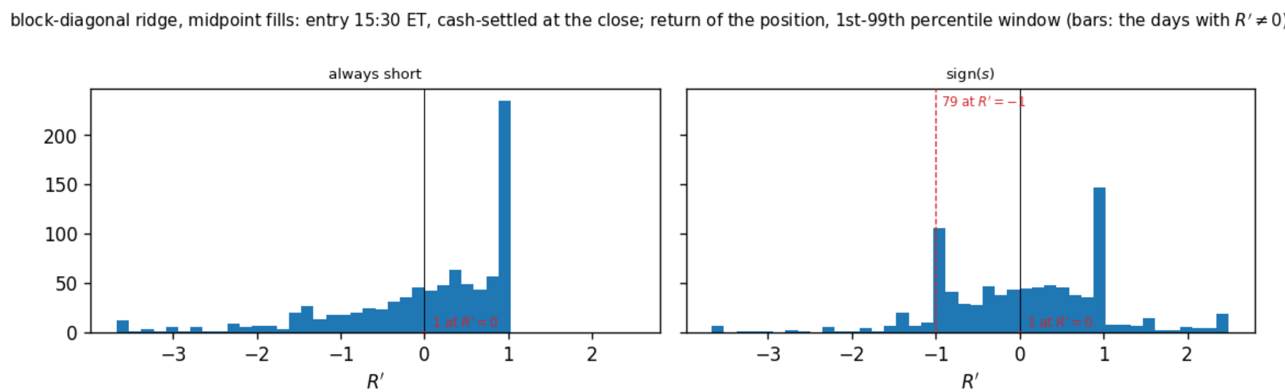
0DTE nearest-OTM package: per-rule return summary, entry 15:30 ET, cash-settled at the close, models compared on the same 866 days

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{ann}}$	$\text{Sharpe}^{\times}_{\text{ann}}$
Short volatility (always short)										
all models (no forecast)	866	0.014	1.128	-10.32	1.00	-2.31	10.89	0.38	0.20	-0.26
sign(s) *(±1 on the sign of s)*										
baseline (HAR + calendar OLS)	866	0.069	1.126	-5.42	10.32	0.58	10.71	1.79	0.97	0.50
block-diagonal ridge	866	0.095	1.124	-5.42	10.32	0.92	10.61	2.48	1.34	0.87
block-diagonal ridge, without the FOMC columns	866	0.090	1.124	-5.42	10.32	1.00	10.59	2.36	1.27	0.80
LightGBM†	866	0.103	1.123	-5.42	10.32	1.01	10.58	2.71	1.46	0.99
XGBoost†	866	0.106	1.123	-5.42	10.32	0.91	10.62	2.78	1.50	1.03
lasso (causally tuned)†	866	0.100	1.123	-6.84	10.32	0.46	10.78	2.62	1.41	0.94
lasso ($\alpha = 10^{-4}$)	866	0.107	1.123	-5.42	10.32	1.22	10.50	2.80	1.51	1.04
elastic net (causally tuned)†	866	0.072	1.125	-6.84	10.32	0.46	10.74	1.90	1.02	0.56



Distribution of the return of the position taken at 15:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window; the mass at −1 is the days the package expires worthless).

One trade a day: enter the nearest-OTM package at 15:30 ET and let it cash-settle at the official close. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean/std}$, with no autocorrelation correction. "ex. kurt" is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. $\text{Sharpe}_{\text{ann}} = (\text{mean/std}) \times \sqrt{252}$; every other column is daily. The one column this table adds to the deck's is $\text{Sharpe}^{\times}_{\text{ann}}$, the same rule's annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck's.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}^2_{\text{hr}} h_t w_t$: the forecast against the implied variance of the same window. On the bars whose vendor implied volatility is a censored solver node the deck's treatment is used here too — the volatility that reproduces the package midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat.